Execution-Constrained Portfolio Rebalancing: Policy Domains, Search and Numerical Reliability
Supplement: formulation, replication and difficulty results

Artur Sepp
LGT

This supplement specifies the methods evaluated with OptimalPortfolios 9.2.0. "Exact Execution" denotes the matched joint solver and "Ranked Execution" the ranked search, historically implemented in `execution/v2`. Method names do not change the identity of the evaluated software. The older operational MIQP implementation has a different policy and is not the matched comparator. This document adds no production rule or numerical tolerance.

1 Declared state domain and formulation

Let $c, m, w \in \mathbb{R}^N$ denote original current, raw target and final weights; k denotes settlement cash. All weights are fractions of net asset value. Table S1 collects the principal symbols. For

Table S1: Notation and distinct numerical conventions.

Symbol	Meaning
c, m, w	Original current, raw target and delivered weight vectors.
q, r	Common prepared base and the funded reference used in coordinate elimination.
N, k	Number of coordinates including cash, and settlement-cash index.
Σ, f, TE	Annual covariance, tracking variance and its square root.
L, U, a, A, b	Hard instrument bounds and fractional group-exposure rows.
$\ell_{ij}, u_{ij}, t_{ij}$	State endpoints and its noncash ticket label.
δ_i, θ_i	Minimum delivered trade and minimum positive holding.
M, B, K	Reserved mandatory names, optional allowance and total ticket capacity.
$10^{-8}, 10^{-7}$	Trade-zero threshold and portfolio audit tolerance, in NAV fractions.
$\epsilon = 10^{-12}$	Search improvement/comparison scale, in annual variance.
10^{-4} bp	Descriptive TE equality band, distinct from audit acceptance.

each noncash instrument i, let J_i be a finite set of labelled closed intervals. State j has endpoints ℓ_{ij}, u_{ij} and ticket label $t_{ij} \in \{0, 1\}$. State construction intersects the permitted movement with the hard instrument bounds, minimum trade and zero-or-minimum-holding rules. The nominal no-trade state is $[c_i, c_i]$ with label zero. Ordinary buy and sell states exclude nonzero movements smaller than the minimum ticket. An eligible full exit may be a separate state.

Mandatory declarations are resolved before search. Their admissible intervals exclude the original holding and have ticket label one. An explicit minimum ticket exemption changes the minimum-size rule only; it does not remove the ticket or the holding floor. The compulsory interval need not be a singleton. Let I_O be the optional instruments and let K, B be the total and optional ticket capacities, when specified. The state domain consists of an admissible labelled interval choice for every noncash instrument, together with

$$\mathbf{1}^\top w = 1, \qquad L \le w \le U, \qquad a \le Aw \le b, \tag{1}$$

and the declared ticket sums. Cash has no ticket and remains bounded.

Proposition 1 (Interval-state formulation)**.** *Introduce binary selectors z_{ij} and contributions v_{ij} satisfying*

$$\sum_{j\in J_i} z_{ij} = 1, \qquad \ell_{ij} z_{ij} \le v_{ij} \le u_{ij} z_{ij}, \qquad w_i = \sum_{j\in J_i} v_{ij}, \tag{2}$$

$$\sum_{i\neq k}\sum_{j\in J_i} t_{ij} z_{ij} \le K, \qquad \sum_{i\in I_O}\sum_{j\in J_i} t_{ij} z_{ij} \le B. \tag{3}$$

Omit a capacity row when that capacity is unspecified. Together with (1), the integer formulation projects exactly onto the declared labelled state domain.

Proof. For each instrument, exactly one selector is one. All other contributions are zero by their two bounds. The remaining contribution is the final weight and lies in its chosen interval. The corresponding labels give precisely the two ticket sums. Conversely, select the declared state of every admissible portfolio, assign its weight to that state's contribution and set all other contributions to zero. Equations (2)–(3) and the unchanged hard rows hold. Cash is determined by funding and retains its bounds. □

This is an application of a standard disjunctive extended formulation; see Vielma (2015), Section 5. It is not a claim that the continuous relaxation of the coupled portfolio problem is ideal. Ideal formulations of individual sets need not describe the convex hull after additional coupling constraints.

Mathematical domain and numerical audit. The proposition concerns the specified intervals and linear rows. Released code permits numerical base-current differences up to the trade-zero threshold and compiles a group residual envelope at the existing audit tolerance. Consequently the proposition is not an equivalence between an exact symbolic domain and every vector that a finite-precision audit may accept. Every candidate is checked against the original supplied inputs. The study keeps audit tolerance 10^{-7} NAV and trade-zero tolerance 10^{-8} unchanged. The descriptive TE equality band 10^{-4} annual bp is not a feasibility tolerance. Search improvement uses 10^{-12} annual variance.

1.1 Funding elimination and objective origin

Fix coordinates that cannot move. Choose a funded reference r and let D contain one column $e_i - e_k$ for every remaining noncash coordinate. Every funded vector with those fixed coordinates has the unique representation $w = r + Dx$, where $x_i = w_i - r_i$. Substitute this expression into all bounds and group rows, including cash bounds. With $\Sigma \succeq 0$ annual covariance,

$$f(r + Dx) = x^\top Q x + 2g^\top x + f(r), \quad Q = D^\top \Sigma D, \quad g = D^\top \Sigma (r - m). \tag{4}$$

For every x, $x^\top Q x = (Dx)^\top \Sigma (Dx) \ge 0$, so Q is positive semidefinite. This is exact funding substitution, not a diagonal approximation.

Multiplication by a positive scalar preserves the mathematical minimiser. Removing $f(r)$ also preserves that minimiser but changes the objective origin seen by a relative stopping rule. Restore both scale and constant when reporting an incumbent or bound. Compare a solver's native bound with a delivered, independently audited incumbent in original variance units. The best native incumbent may itself fail the audit; its native gap can then be smaller than the delivered portfolio's gap. Restricted Kernel Search bounds are not global bounds for the full execution problem.

1.2 Perspective guidance

Choose a nonnegative diagonal matrix $\operatorname{diag}(d)$ such that $Q_0 = Q - \operatorname{diag}(d) \succeq 0$. Let y_i be optional ticket occupancy; use $y_i = 1$ for mandatory coordinates. The no-ticket state fixes $x_i = 0$. The perspective objective in unscaled funded coordinates is

$$f(r) + 2g^\top x + x^\top Q_0 x + \sum_i d_i\, \psi(x_i, y_i), \qquad \psi(x,y) = \begin{cases} x^2/y & y > 0, \\ 0 & x = y = 0, \\ +\infty & y = 0,\ x \neq 0. \end{cases}$$

Each perspective term is convex. At integer occupancy it equals x_i^2 on every admissible state, preserving the original objective. For $0 < y_i \leq 1$ it is at least x_i^2, so it strengthens the ordinary continuous objective on the same relaxed state rows. The resulting optimal relaxation value is a lower bound on the integer optimum. These are standard perspective properties (Frangioni and Gentile, 2006; Günlük and Linderoth, 2010).

Released guidance uses a conservative verified diagonal split, not an optimised diagonal decomposition; a zero split gives no strengthening. An approximate conic primal value is not automatically a lower bound. The separate OP bound routine constructs a supporting plane and minimises it over the linear relaxation, with an LP dual residual correction on bounded variables. This certificate route is separate from native mixed-integer bound diagnostics. The prospective Exact arms retain the released non-perspective default; Kernel and Ranked use perspective guidance for priorities. The study therefore evaluates those declared implementations, not the strongest possible mixed-integer formulation.

1.3 A corridor counterexample

Let

$$c = (0.2, 0.2, 0.6), \quad m = (0.3, 0.3, 0.4), \quad \Sigma = 0.01 \begin{pmatrix} 1 & 0.9 & 0 \\ 0.9 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

with cash last, hard bounds $[0, 1]$, one noncash ticket, no mandatory action, minimum trade 0.005 and holding floor 0.0025. The ordinary risky corridors are $[0.2, 0.3]$ and cash remains in $[0, 1]$. If the first instrument trades, the second remains at 0.2. Writing $x = w_1 - 0.3$ gives

$$f(w) = 0.01(x^2 - 0.18x + 0.01).$$

The unrestricted minimum is at $x = 0.09$, giving $w = (0.39, 0.2, 0.41)$, variance 0.000019 and TE 43.5889894354 annual bp. In the corridor, $x \in [-0.1, 0]$ and the minimum is at zero, giving $w = (0.3, 0.2, 0.5)$, variance 0.0001 and TE 100 bp. Trading the second instrument gives the symmetric solutions; no trade is worse. These scalar cases exhaust the ticket choices independently of either execution algorithm.

The overshoot offsets an untraded correlated underweight. Thus a corridor is an economic policy restriction, not a generally lossless reduction of the full hard domain. The replication test checks the scalar objectives and both solver domains without changing either audit tolerance. Comparisons of time-limited Full and Corridor incumbents do not by themselves measure the optimal restriction cost: their search errors can differ.

2 Ranked Execution: ordered specification

A support S maps selected optional instruments to executable state indices. Mandatory state assignments are separate. The released defaults are rescue budget $R = 24$, mandatory-assignment budget $H = 32$, improvement-pass budget $P = 3$ and variance-improvement threshold $\epsilon = 10^{-12}$.

Identifier and then state index resolve priority ties. A returned incumbent must pass the original-input audit; a relaxed or tentative vector never serves as a delivery solely because it produced a promising priority.

2.1 Sizing and common state

Maintain a support cache keyed by sorted (i, j) pairs. For a previously unseen support, SIZE(S) first checks capacity, then tries at most H mandatory assignments in distance-from-base and state-index order. Pin unselected optional names to their no-ticket states and jointly size all selected names with the common convex quadratic model and unchanged hard rows. Audit each candidate and cache the best accepted result, or the attempted failure. Different search branches have separate visited sets but share this cache. A failed sizing attempt is not a proof that the state assignment is infeasible. Candidate acceptance requires an improvement greater than ϵ.

The funded scalar priority for an optional movement d in name i is

$$f(w + d(e_i - e_k)) - f(w) = 2d(e_i - e_k)^\top \Sigma(w - m) + d^2(e_i - e_k)^\top \Sigma(e_i - e_k).$$

Minimise this quadratic over the state's movement interval; for two-name priorities retain the cross term between their funded directions. A priority does not replace the subsequent joint sizing or audit.

2.2 Search order

The following order matters because later seeds, caches and incumbents depend on earlier accepted results.

1. Compile states, retain the audited prepared base and determine optional capacity. If the declared state domain conflicts with capacity, retain an audited base if available. Enumerate the bounded mandatory assignments once.

2. From the base, greedily add distinct optional names using the funded scalar priorities. Update the covariance product after each tentative move. Stop at capacity or when no scalar gain remains.

3. Size the empty support when mandatory intervals can move, then the greedy support. For failed sizing, try up to R drop, admission, state-change or replacement rescue supports ordered by coupled group pressure and identifier.

4. Run single-drop and best-ranked replacement passes. Run up to P coordinated anchors sharing R shortlisted support trials. Initial coordinated moves edit one or two instrument states and use conditional funded quadratics.

5. When capacity is positive and below the candidate count, solve the unrestricted perspective guidance relaxation. Select at most capacity names by fractional occupancy, with their largest eligible state selectors. Explore this seed separately while preserving the global incumbent. Run expanded coordinated improvement, complete single-replacement sweeps and up to R shared coordinated trials, including the guidance seed.

6. Restart from the completed incumbent and a distinct unrestricted guided seed. Exclude each of up to R incumbent names in separate guidance problems. Give every distinct seed at most P complete one-for-one replacement sweeps. An inferior audited seed may initiate a branch, but only its completed improvement may replace the global incumbent. Failed rounded seeds may receive a one-swap rescue.

7. Form the identifier-ordered union of incumbent and unrestricted-guidance states. Try inclusion then exclusion, up to R conditional guidance calls. Guidance changes the seed only: sizing always uses the unmodified problem. Restart distinct seeds and revisit the completed conditional incumbent.

8. Complete up to P expanded anchors sharing R^2 coordinated trials, skipping cached supports. Expanded moves include two admissions with one or two removals. Apply a necessary interval-envelope funding and group screen before sizing. Each anchor also performs a complete replacement sweep.

9. Sort cached audited alternatives by objective and identifier and restart at most R. The first branch improving the global incumbent receives one further P-anchor, R^2-trial completion. Return the best audited incumbent with truncation and failure diagnostics.

An *anchor* is the incumbent support at the start of an improvement pass. *Rescue* seeks feasible sizing after a tentative support fails. A *shortlist* limits supports sent to sizing after priority enumeration. *Completion* denotes a finite extra neighbourhood budget, not exhaustive search. A bridge restart starts from a cached alternative while preserving the best incumbent in separate state. None of these stopping conditions certifies global optimality.

2.3 Computational scope

Let C be eligible optional names, S_t ticket states, B capacity, G group rows and T distinct supports submitted to sizing. At most HT fixed-state QPs are attempted, in addition to guidance solves. A complete replacement sweep examines order BS_t proposals. Two admissions with up to two removals can generate order $(B + B^2)S_t^2$ proposals per anchor before the bounded shortlist. Screens add dependence on G and some full-vector work. A bounded sizing budget therefore does not establish an $O(NB)$ runtime or a small wall-clock bound.

An exact factor-plus-diagonal covariance with r factors uses order $Nr+r^2+N$ input storage. The released implementation nevertheless constructs a dense $N \times d$ funded map and a dense $d \times d$ funded Hessian, where d counts unfixed noncash coordinates. Mandatory intervals can contribute to d. QIS reporting also materialises the $N \times N$ covariance. Reported peak worker memory includes imports and reporting; it is not a measurement of only the branch-and-bound model. Structured input does not establish linear-memory execution.

3 Kernel Search comparison

The comparator adapts the restricted-kernel and bucket scheme in Guastaroba and Speranza (2012, 2014) to the same convex quadratic execution states. It does not reproduce their index-tracking MILP. It ranks names by fractional ticket occupancy from the perspective relaxation, rather than LP reduced costs. The initial kernel has $\min(C, \max(24, 2B))$ names, with stable identifier ties. Divide the remaining ordered names into disjoint buckets of that initial size.

First solve on the kernel, then make one pass through kernel-plus-bucket subproblems. Pin excluded optional names to the prepared base. Retain every mandatory state and all original hard rows. Each bucket subproblem requires at least one bucket ticket and has an incumbent cutoff of best variance plus 10^{-12}. Divide the remaining internal allowance among remaining stages. Names selected by an audited bucket solution join the kernel. There is no ageing, removal or later refinement. The common prepared base supplies a feasible initial candidate when it passes audit; no incumbent is imported from another comparison arm.

The 60-second allowance includes guidance, model construction and subproblem search. Independently audit both original and restricted domains and verify bucket use before retention. Restricted-subproblem bounds cannot certify the unrestricted problem. The adaptation is a

literature-based control with declared choices, not a claim of the best attainable Kernel Search tuning.

4 Prospective difficulty protocol

The machine-readable authority is `study_protocol.json`. Before any evaluation solve, the runner freezes all cases, source bytes, wheel identity, ordering, versions and resource limits. The retained earlier 313 cases and the four resource-planning pilot cases remain separate from this evaluation. No solver is modified after inspecting this cohort.

4.1 Sparse synthetic books

There are 18 scaling cases: 300, 1,000 or 3,000 noncash universe names; 10, 30 or 100 optional tickets; and two bundled profiles. Current holdings use 96 names, each at least 0.6%, with 6% settlement cash. Targets add 100 names and hold 2% cash. The 0.5% minimum ticket and 0.25% holding floor are fixed. Universe size, held positions and executable candidates are distinct: the wide cells have 155 eligible optional names and the tight cells have 190. The number of ticket states ranges from 200 to 280 across these cells.

The wide profile combines disjoint groups, diffuse target support and two mandatory exits. The tight profile combines overlapping fractional rows, exit-rich targets and six mandatory exits. Each profile has one predeclared seed, shared across its size and capacity cells. Risk uses exactly twelve factors plus diagonal residual variance. Separate random streams construct current holdings, target and risk; streams also depend on universe size. Within a size/profile cell, changing ticket capacity preserves those inputs. Cross-size differences therefore include different random books and cannot isolate a pure universe-size effect. Profiles bundle multiple features and differ in seed: their contrast is not a factorial estimate or population superiority claim. An independently audited base witnesses feasibility.

4.2 External covariance adaptation

Four further cases use the OR-Library `indtrack1` and `indtrack6` price panels, with 31 and 457 stocks, at observation cuts 145 and 290. Holdings, targets, cash and every execution constraint remain synthetic. The source format provides price observations without calendar dates. Artificial weekly labels and 52 periods per year are explicit model conventions, not recovered historical dates or frequencies.

QIS computes point-in-time demeaned log-return EWMA covariance with span 52, using only prices through the cut. Annualisation multiplies the period covariance by 52. Synthetic cash variance is 10^{-6}. The source index series is not used as the execution target. These are externally sourced risk cases, not replications of the original index-tracking objective. Public code records owner URLs, source checksums and transformation. Raw price files and their derived covariance arrays are excluded from the redistribution archive pending established reuse terms; fetch from the owner to reproduce locally.

4.3 Accounting and denominators

The 23 cases comprise the 18 scaling cells, four external-risk cases and the analytic counterexample. Each has independent Exact60, Ranked and Kernel60 arms. Each tight, 30-ticket scaling cell additionally has Exact10, Exact600, Dense10 and Full60; the counterexample also has Full60. This totals 82 planned deliveries. Dense10 changes covariance representation only; Full60 changes the corridor policy domain while preserving common preparation and hard rows.

Cases run serially with one solver and numerical-library thread. Arm order rotates by case index, each in a fresh process. Preparation has a 30-second internal allowance and a 90-second

process ceiling. Exact10, Exact60 and Exact600 receive 10, 60 and 600 seconds, with process ceilings 60, 120 and 660 seconds. Kernel60, Dense10 and Full60 have 120-second process ceilings. Ranked retains its released stopping controls with a 180-second external process ceiling; this does not add a new internal stopping algorithm.

Charge common preparation to every arm, time covariance conversion separately, and record engine, end-to-end worker and complete process times. A killed worker without a delivered result remains a failed delivery, even if it had an internal incumbent. Preserve any boundary result separately. Timing summaries include failures where a process duration exists; missing memory measurements are not invented. Paired TE comparisons state the number of audited pairs and the number missing. Native statuses and bounds, audit acceptance and independently proved optimality remain separate evidence.

The supplied prepared base is already feasible. Time to the first audited native candidate can therefore measure acceptance of that lifted start rather than discovery of an improvement. Kernel likewise records initial feasibility at zero when the base passes audit. Ranked has no new anytime-delivery interface in this study: a process ceiling means that no portfolio was returned by that deadline, not that the search never held a feasible internal incumbent. The different declared resource controls compare these released interfaces; they are not a matched wall-clock anytime experiment.

Source and input identity do not imply identical time-limited search paths on different hosts or reruns. A wall-clock cutoff can stop at a different incumbent. The resource protocol, all failed attempts and original-input audits are part of reproduction; the retained run provides the exact observations used here.

5 Isolated numerical mechanism controls

These controls use retained public regression examples, separately from the prospective instances. They change equivalent graph representations in an isolated research process; installed package files and production settings remain unchanged. Their purpose is to expose mechanisms and their limits, not to provide additional out-of-sample success counts.

Objective translation. For an incumbent value F_U, lower bound F_L and positive scale α, translation to $\alpha f + \beta$ leaves the mathematical minimiser unchanged but changes the relative gap, with denominator floor $\eta > 0$, to

$$\frac{\alpha|F_U - F_L|}{\max(\eta, |\alpha F_U + \beta|)}.$$

Take $F_U = 10^{-10}$, $F_L = 9 \times 10^{-11}$ and remove a constant 10^{-3}. The original relative gap is 0.1; the translated gap is approximately 10^{-8}. The absolute gap 10^{-11} exceeds the unchanged 10^{-12} variance allowance. Thus a 10^{-4} relative criterion can accept the translated pair while rejecting the original pair. This is an analytical bound-pair example, not fabricated native solver output.

Equivalent quadratic cones. Use the same public 24- and 72-optional-name examples and the same state rows. Compare $x^\top Q x/10^4$ with $(x/100)^\top Q(x/100)$, where x is funded displacement in basis points. Retain the linear term, constant and original variance origin in both graphs. Both variants receive ten seconds, one native thread and the released numerical settings. Every returned candidate passes through the same original-input audit before it may displace the prepared base. For the 24-name example an independent full-weight reference on a known support provides a feasible witness. It is not automatically a global certificate for the 72-name problem. A control that does not reproduce a prior defect under this runtime is reported as such.

Satisfied constant rows. After fixed-coordinate elimination, a group exposure can become constant. The released sizing compiler removes a row only when its coefficients are exactly zero and its constant satisfies both compiled endpoints. It retains every nonconstant row and every constant contradiction. The control compares retaining versus removing satisfied rows, for both a lower and an upper fractional-exposure limit. Residuals are $(1 - 10^{-8}) \times 10^{-7}$ and 1.01×10^{-7} NAV, respectively inside and outside the unchanged audit envelope. This gives eight graph variants. The feasible cases are checked against an independent active-set calculation after removing the mathematically redundant row; the outside-envelope cases have an immediate constant-row infeasibility witness. Native status, errors and audited delivery are retained separately. Removing a redundant row is not permission to discard a genuine constraint breach.

6 Completed prospective results

All 23 cases prepare successfully and all 82 declared attempts are retained. There are 64 audited deliveries and 18 process-ceiling failures, all from Ranked. No returned portfolio fails the independent original-input re-audit. Preparation recognises an already audited base in every new case; this sample therefore does not measure the difficulty of discovering compulsory repairs. It tests optional search conditional on the declared mandatory movements and common feasible base.

6.1 Delivery, quality and computation

Exact60 and Kernel60 deliver in all 18 scaling cases. Ranked delivers in two and reaches its 180-second process ceiling in 16. Of the two returned Ranked portfolios, one ties Exact60 and one improves it by 2.402490 bp; both have higher TE than Kernel60. Kernel60 improves on Exact60 in 11 cases, ties in four and is worse in three. Its signed differences range from −55.718876 to 23.386397 bp. These are comparisons of audited incumbents, not errors relative to a proven optimum. The fixed resource controls and different delivery interfaces are part of the comparison.

Table S2: Delivery and process cost in the prospective study. Exact and Kernel have 60-second internal allowances; Ranked retains released defaults with a 180-second process ceiling. Process time includes imports and reporting. Memory is the largest observed worker peak, with its observation count in parentheses; killed workers have no peak measurement.

Sample	Method	Audited	Median s	95th pct. s	Observed MiB
Scaling	Exact	18/18	66.312	67.411	746.137 (18)
Scaling	Ranked	2/18	180.019	180.026	497.750 (2)
Scaling	Kernel	18/18	66.647	67.823	762.730 (18)
External risk	Exact	4/4	35.760	65.471	321.340 (4)
External risk	Ranked	2/4	99.137	180.021	271.801 (2)
External risk	Kernel	4/4	35.840	66.089	348.336 (4)

The four external-risk cases give a different result. Exact60 and Kernel60 both deliver four portfolios. They tie on the two 31-name cases; Exact has lower TE on the two 457-name cases, by 0.191694 and 0.361215 bp. Ranked ties Exact on the two smaller cases and returns no portfolio by its deadline on the two larger ones. Neither the scaling nor external sample supports an unconditional ranking of algorithm families.

The optional ticket cap binds in ten of 18 scaling Exact60 deliveries and 12 of 18 Kernel60 deliveries. The two completed scaling Ranked calls use zero and two optional tickets. Current holdings, target support and executable candidate counts are distinct from the nominal universe. In particular, increasing the universe to 3,000 does not create 3,000 Corridor candidates.

Table S3: Paired audited TE differences, left method minus Exact60, in annual bp. Negative differences favour the left method. Open Exact gaps mean these are incumbent comparisons. The equality band is 10^{-4} bp.

Sample	Left	Paired	Lower	Ties	Higher	Min bp	Max bp
Scaling	Ranked	2/18	1	1	0	-2.402	0.000
Scaling	Kernel	18/18	11	4	3	-55.719	23.386
External risk	Ranked	2/4	0	2	0	0.000	0.000
External risk	Kernel	4/4	0	2	2	0.000	0.361

Table S2 reports complete method-process duration, including imports and reporting, and keeps killed calls in the timing denominator. Adding each common preparation process gives scaling median research-pipeline times of 72.871 seconds for Exact, 186.578 for Ranked and 73.245 for Kernel. These research-process figures include two import phases; they are not direct predictions for a persistent production process. The largest observed worker peaks are approximately 746 MiB for Exact and 763 MiB for Kernel. The worker includes dense QIS risk reporting. Missing killed-worker memory measurements do not imply low memory consumption by Ranked.

6.2 Longer budgets, domains and native diagnostics

Only two of the 18 scaling Exact60 calls report native integer-optimal status; the other 16 report primal-feasible status. Their median native relative gap is 15.771%, and the maximum is 72.637%. Two external-risk Exact60 calls report integer-optimal status and two primal-feasible status. There is no native-bound contradiction against an available audited witness in the same domain at the unchanged 10^{-12} variance comparison scale. This check does not independently certify an unfinished bound or optimum.

Table S4: All prespecified 30-ticket, tight-profile controls. TE is annual bp. Full60 changes the policy domain; its difference is not search error on the Corridor problem. The final column reports the native relative gap for Exact600, which can differ from the gap of the audited delivery.

Names	Exact10	Exact60	Exact600	Dense10	Full60	Native gap %
300	133.627	92.849	90.613	133.627	133.627	7.554
1000	159.813	159.813	159.813	159.813	159.813	61.837
3000	210.198	210.198	112.441	210.198	210.198	3.170

The three Exact600 controls remain unresolved. Increasing 60 seconds to 600 improves the 300-name case by 2.236112 bp and the 3,000-name case by 97.757191 bp. The 1,000-name case remains at its common prepared incumbent, with native relative gap 61.837%; Kernel60 improves that incumbent by 55.718876 bp. Native callback records for that Exact600 case contain no rejected better vector. Its failure to improve cannot be attributed to rejection at an audit boundary.

Across the scaling Exact60 calls, 20 native candidate events fail the audit; these are candidate events, not 20 failed portfolio deliveries. A native incumbent can differ from the accepted portfolio. For example, the 300-name, ten-ticket wide case has native incumbent variance 0.000130181637 but delivers 0.000132001026 after the better native candidate exceeds the group audit envelope. Both the native and delivered gaps are retained, and no tolerance is widened to accept that candidate.

Dense10 and structured Exact10 return the same prepared base in all three representation controls. This result does not show that their search paths or memory costs are equal. Likewise all three Full60 controls return the base: one is worse than Corridor Exact60 and two tie. Those

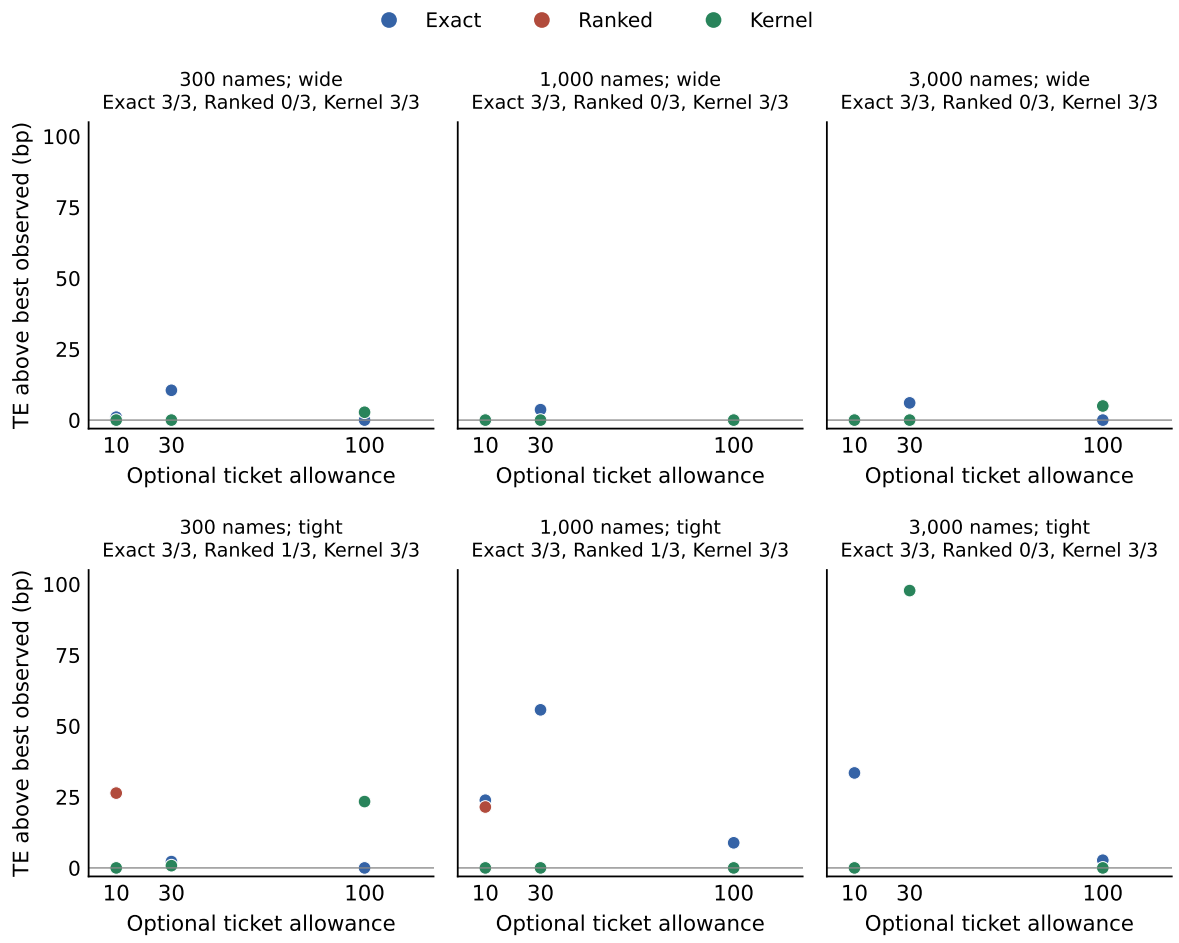


Figure 1: Audited TE above the best observed Corridor delivery in each scaling case. The reference includes Exact600 where it was prespecified. Full-domain portfolios are excluded. Panel counts retain missing deliveries; no point is imputed for a process ceiling. Zero means agreement with an observed incumbent, not a certificate of optimality.

unfinished results cannot estimate the optimal cost of the corridor restriction. The analytic case does establish a possible cost independently: all three Corridor methods give 100 bp, while Full gives 43.588989 bp, matching the scalar calculation.

6.3 Historical controls and interpretation

The earlier 313-case cohort remains a separate immutable study. Its ETF decisions have only zero to five effective optional candidates; its four fresh-holdout Ranked misses and 12 economically tiny ETF advantages remain visible in Table S5. Its primary/fallback wrapper had zero natural fallbacks and nine successful injected fallbacks. No wrapper was run on the new difficulty sample, so standalone failures cannot be converted into an observed fallback frequency.

Table S5: Retained 313-case cohort: Ranked minus independent Exact TE, in annual bp. Negative differences favour Ranked. The equality band is 10^{-4} bp; the smaller ETF differences remain visible.

Sample	Paired	Ranked lower	Ties	Ranked higher	Minimum	Maximum
ETF	219	12	207	0	-0.000941	0.000043
Existing synthetic	26	0	26	0	0.000000	0.000000
Fresh holdout	68	0	64	4	0.000000	2.457878

These observations support a study of the declared execution domain, numerical representation and measured computational limits. They do not support a claim that Ranked is generally superior to Exact, that Exact is universally faster or better, or that a fallback wrapper removes search risk. Kernel's advantage is substantial in several synthetic cells but reverses in others. The prespecified bundled profiles, limited seeds and conditional supplied-input design do not establish population or realised investment-performance claims. Any formulation strengthening or new stopping interface requires separate development and another frozen evaluation.

6.4 Numerical mechanism outcomes

All four cone controls return audited portfolios and native integer-optimal labels. Nevertheless the unbalanced representations report bounds above better audited witnesses on the same inputs, by 7.630×10^{-11} and 5.439×10^{-10} original variance for 24 and 72 optional names. Both exceed the unchanged 10^{-12} comparison scale. The balanced variants show no such contradiction at that scale. The resulting TE differences are economically small; the issue is the numerical validity and attribution of the bound. These isolated research variants do not alter released code.

Table S6: Equivalent cone controls under unchanged native settings. Variances are in original annual units. Excess is the native bound minus the best audited witness from the two equivalent graphs, in units of 10^{-12} variance. A native optimal label does not remove a contradictory witness.

Optional	Graph	Delivered variance	Native bound	Excess
24	Unbalanced	0.000176888072	0.000176888082	76.299
24	Balanced	0.000176888006	0.000176888006	0.000
72	Unbalanced	0.000164967505	0.000164967562	543.918
72	Balanced	0.000164967018	0.000164967018	0.000

The constant-row controls do not reproduce the historical sizing failure in this runtime. Both retained-row and reduced-row graphs solve the two inside-envelope cases and agree with the independent active-set variance 0.0013375 to floating-point precision. All four outside-envelope graph variants are infeasible. Thus these controls support preservation of genuine breaches, but

provide no measured delivery advantage for row elimination here. The objective-translation example remains an analytical stopping-rule counterexample, separately labelled from native observations.

7 References

- Frangioni, A. and Gentile, C. (2006). Perspective cuts for a class of convex 0–1 mixed integer programs. *Mathematical Programming* 106(2), 225–236. https://doi.org/10.1007/s10107-005-0594-3.

- Günlük, O. and Linderoth, J. (2010). Perspective reformulations of mixed integer nonlinear programs with indicator variables. *Mathematical Programming* 124, 183–205. https://doi.org/10.1007/s10107-010-0360-z.

- Guastaroba, G. and Speranza, M.G. (2012). Kernel Search: An application to the index tracking problem. *European Journal of Operational Research* 217, 54–68. https://doi.org/10.1016/j.ejor.2011.09.004.

- Guastaroba, G. and Speranza, M.G. (2014). A heuristic for BILP problems: The Single Source Capacitated Facility Location Problem. *European Journal of Operational Research* 238, 438–450. https://doi.org/10.1016/j.ejor.2014.04.007.

- Vielma, J.P. (2015). Mixed integer linear programming formulation techniques. *SIAM Review* 57(1), 3–57. https://doi.org/10.1137/130915303.

- Beasley, J.E., Meade, N. and Chang, T.-J. (2003). An evolutionary heuristic for the index tracking problem. *European Journal of Operational Research* 148, 621–643. https://doi.org/10.1016/S0377-2217(02)00425-3.

- Canakgoz, N.A. and Beasley, J.E. (2009). Mixed-integer programming approaches for index tracking and enhanced indexation. *European Journal of Operational Research* 196, 384–399. https://doi.org/10.1016/j.ejor.2008.03.015.

- OR-Library. Index tracking instances and format. https://people.brunel.ac.uk/~mastjjb/jeb/orlib/indtrackinfo.html.

- Sepp, A. OptimalPortfolios: portfolio construction software. https://github.com/ArturSepp/OptimalPortfolios.

- Sepp, A. QuantInvestStrats (QIS): quantitative investment analytics. https://github.com/ArturSepp/QuantInvestStrats.